

Cross-Cultural Validation and Ideological Fairness of a Historical Perspective Taking Instrument: Evidence from Czech Secondary Students

Juda Kaleta

2026

Juda Kaleta

Institute of History, Faculty of Arts, Charles University, Czech Republic

Email: juda.kaleta@ff.cuni.cz | ORCID: 0000-0003-0043-4039

Abstract

This study reports ~~the first~~ a Czech adaptation of the Hartmann and Hasselhorn Historical Perspective Taking (HPT) instrument and examines whether its politically charged scenario introduces construct-irrelevant variance (CIV) related to students' ideological attitudes. Among 293 Czech secondary students, confirmatory factor analysis ~~(CFA)~~ confirmed the showed good fit for a correlated three-factor ~~structure.~~ Item Response Theory (IRT)-based differential item functioning (DIF) representation compatible with the Czech data. An item-level bias analysis detected no ~~biased items~~ differential item functioning across ideology groups with this sample and procedure. Multi-group ~~CFA~~ demonstrated full scalar measurement invariance factor analysis produced fit-index

changes compatible with scalar invariance, but small extreme groups and unmodeled clustering limit the strength of that evidence. Historical knowledge ~~predicted HPT scores;~~ ~~ideology was unrelated~~ correlated with the primary six-item HPT composite ($r = .34$, 95% CI [.23, .43]); right-authoritarian attitudes were unrelated ($r = -.05$, 95% CI [-.16, .07]). These findings support a ~~transferable~~ possible protocol for evaluating measurement fairness when assessment content carries attitudinal valence, subject to the limits of this single scenario, item format, and ideologically restricted sample.

Keywords: historical perspective taking, cross-cultural validation, measurement invariance, differential item functioning, construct-irrelevant variance, test fairness

1. Introduction

Cross-cultural adaptation of educational and psychological instruments requires more than linguistic translation: it demands evidence that the adapted version measures the same construct, with equivalent psychometric properties, in the target population (American Educational Research Association, American Psychological Association, & National Council on Measurement in Education, 2014; Hambleton, Merenda, & Spielberger, 2005; International Test Commission, 2017). When assessment content carries attitudinal valence—when scenarios, stimuli, or items touch on beliefs that respondents may hold—an additional fairness question arises: does the content introduce construct-irrelevant variance (CIV) by functioning differently for subgroups whose attitudes align with that content (Messick, 1995)? This question is especially pressing for scenario-based instruments in the social sciences, where rich contextual material is needed to elicit

40 complex reasoning but may simultaneously activate respondents' pre-existing attitudes.
41 Historical perspective taking (HPT)—the capacity to reconstruct the beliefs, values, and
42 circumstances that made past actors' decisions intelligible in their own time (Hartmann
43 & Hasselhorn, 2008; Lee & Ashby, 2001)—provides a particularly informative case. HPT
44 instruments routinely use politically and morally charged historical scenarios as the
45 stimulus for disciplinary reasoning (reasoning according to the methodological norms of
46 the historical discipline, such as source criticism and ~~contextualisation~~contextualization;
47 van Drie & van Boxtel, 2008; Seixas & Morton, 2013; Wineburg, 2001).

48 The instrument ~~developed by is among the most extensively validated standardized~~
49 ~~HPTmeasures in European research. Originally reported by~~ Hartmann and Hassel-
50 horn (2008) provides a scenario-based measure of HPT. It was validated with German
51 ~~secondary students, it has been adapted for use in the Netherlands and employed in~~
52 ~~comparative studies of historical reasoning~~Grade 10 students (Hartmann & Hasselhorn,
53 2008) and subsequently translated and studied with Dutch students aged 10–17 (Huijgen,
54 van Boxtel, van de Grift, & Holthuis, 2014, 2017). This evidence base is promising but
55 remains concentrated in two language contexts and a small number of studies. The
56 instrument operationalizes HPT through a scenario-based approach in which students
57 read about a historically situated dilemma and ~~respond to nine Likert-scaled items~~
58 ~~capturing three developmental levels: presentism/othering of people~~ rate how well
59 nine explanations fit the situation on a four-point response scale in three interpretive
60 response categories: present-oriented perspective (POP), ~~recognition of the agent's social~~
61 ~~role and situational context~~ role of the historical agent (ROA), and historical contex-

tualization (CONT). ~~This structure reflects a theoretical progression from naive POP~~
~~and CONT represent the clearest contrast between~~ present-centered judgment ~~toward~~
~~increasingly sophisticated historical reasoning and contextual reasoning; ROA's position~~
~~as an intermediate level remains theoretically less certain~~ (Hartmann & Hasselhorn, 2008;
Lee & Ashby, 2001).

The present study reports ~~the first a~~ Czech adaptation and psychometric evaluation of the
Hartmann and Hasselhorn HPT instrument. Two considerations motivate this work. First,
no validated Czech-language HPT instrument ~~currently exists, despite growing interest~~
~~in historical thinking assessment within Czech education reform . The Czech adaptation~~
~~addresses this gap and provides the research community with a tool for cross-cultural~~
~~comparisons of HPT development~~ was identified. Czech initiatives have developed digital
historical-literacy tasks and measures of epistemic beliefs about history (Činátl, Najbert, &
Ripka, 2021; Ripka, Sýkorová, Münich, & Chvojka, 2024); ~~the present adaptation provides~~
~~a candidate instrument for further Czech validation and future cross-national equivalence~~
~~studies.~~

Second, the instrument's scenario content raises a specific fairness question. ~~The scenario~~
~~describes a young man in the Weimar Republic deliberating whether to join an extremist~~
~~political movement. Items that require students to contextualize this figure's reasoning~~
~~Hannes discusses Germany's crisis and the 1930 election; students judge how well nine~~
~~explanations fit his situation, including whether he could vote for an anti-democratic~~
~~party such as the NSDAP. These judgments~~ could, in principle, function differently
for students whose political attitudes align with the scenario's ideological content.

Students with right-authoritarian sympathies might find it cognitively easier to endorse contextualizing responses—not because they are reasoning historically, but because the responses resonate with their own attitudes. If present, such a pattern would constitute construct-irrelevant variance (CIV): systematic score variation attributable to a factor extraneous to the target construct (American Educational Research Association et al., 2014; Messick, 1995).

This concern is not merely theoretical. Research on identity-protective cognition demonstrates that individuals process information through the lens of pre-existing beliefs (Jost, Glaser, Kruglanski, & Sulloway, 2003; Kahan, Peters, Dawson, & Slovic, 2017). Recent work has documented how ideological content can infiltrate competence judgments in history assessment, whether through teacher scoring bias in open-ended narrative assessment (Alvén, 2019) or through construct comparability challenges in cross-cultural settings (Badham, Meadows, & Baird, 2025). The question central to the present study is whether the HPT instrument’s psychometric properties—its factor structure, item parameters, and score interpretations—hold equivalently across students with different political attitudes, or whether politically charged content introduces measurement bias.

The study contributes to the literature in three ways: (a) it ~~demonstrates a transferable~~ applies a possible methodological approach for evaluating whether attitudinally charged assessment content introduces construct-irrelevant variance, (b) it provides DIF and ~~measurement invariance evidence for a fairness-relevant grouping variable—political ideology—that has not previously been examined in HPT measurement~~ measurement invariance evidence across political-ideology groups, and (c) it ~~reports the first published Czech~~

~~validation of the Hartmann and Hasselhorn HPT instrument, extending the~~ adds Czech
evidence to the small cross-cultural evidence base for this measure. To establish this
evidence, we conducted confirmatory factor analysis (CFA), reliability analysis, intra-
class correlation estimation, IRT-based differential item functioning (DIF) analysis, and
multi-group CFA for measurement invariance testing.

2. Literature Review

2.1 Historical Perspective Taking and Its Measurement

HPT refers to the ability to understand past actors' decisions by reconstructing the histor-
ical context in which they operated (Hartmann & Hasselhorn, 2008; Lee & Ashby, 2001).
HPT is a second-order (procedural or disciplinary) concept—as distinct from first-order
(substantive) concepts such as “revolution” or “empire,” it concerns the meta-level reason-
ing processes historians use to make sense of the past (Seixas & Peck, 2004). The concept
has roots in British traditions that progressively refined the construct from historical empa-
thy (*Children's concepts of empathy and understanding in history*, 1987; Shemilt, 1984) toward
rational reconstruction (Lee & Ashby, 2001). It has been situated within broader frame-
works of historical reasoning that encompass evidence use, argumentation, and substan-
tive concept deployment—the application of domain-specific content knowledge about
events, causes, and actors (van Boxtel & van Drie, 2018; van Drie & van Boxtel, 2008;
VanSledright, 2014). The literature variously labels this capacity “historical empathy,”
“perspective taking,” or “rational reconstruction.” We follow Hartmann and Hasselhorn
(2008) in adopting the term “perspective taking” within the rationalist tradition of ~~;~~ **which**

~~emphasises~~ Lee and Ashby (2001), which emphasizes cognitive reconstruction of context rather than affective identification with past actors (see Barton & Levstik, 2004, for a contrasting view that ~~emphasises~~ emphasizes affective engagement alongside cognitive reconstruction; see also ~~, for a more recent treatment~~ Endacott, 2014; Endacott & Brooks, 2013, for treatments of the empathy/perspective-taking distinction). The present instrument therefore operationalizes cognitive perspective taking, not the fuller cognitive-affective construct of historical empathy. Across these frameworks, HPT is understood as a cognitively demanding, knowledge-dependent competence—not a disposition but a disciplinary achievement requiring both domain-specific knowledge and the capacity to deploy that knowledge in reconstructing past perspectives (Chapman, 2021; VanSledright, 2014; Wineburg, 2001). Distinguishing genuine disciplinary reasoning from attitude-driven responding is a measurement challenge that extends beyond HPT to epistemic cognition in history more broadly (Maggioni, VanSledright, & Alexander, 2009).

~~operationalised HPT into a developmental model with three levels:~~ Hartmann and Hasselhorn (2008) operationalized HPT using three response categories: ~~presentism~~ present-oriented perspective (POP), where students judge historical actors by contemporary standards; ~~recognition-role of the historical agent's-role~~ (ROA), where students ~~acknowledge the agent's social position without full contextual reconstruction~~ explain action through familiar or stereotyped social roles; and *contextualization* (CONT), where students situate decisions within the specific historical conditions that rendered them intelligible. ~~This developmental progression from present-centred judgment toward contextualisation~~ The POP-CONT continuum from present-centered judgment toward contextualization has

empirical parallels in earlier work on students' historical understanding (Lee & Shemilt, 2003). The original published validation study ($n = 170$ German students, Grade 10) ~~recovered three levels through principal component analysis~~; Hartmann & Hasselhorn, 2008) placed raw POP and CONT at opposing poles of one principal component and ROA on a second, less clearly interpreted component (explained variance = 51%), ~~and students classified at the highest level~~; a separate latent-class analysis identified three student classes. Students classified in the highest class had significantly better history grades; however, PCA is an exploratory technique, and the present study ~~provides the first confirmatory factor analytic test of the three-level structure~~. The instrument was ~~extended by~~ tests a correlated three-factor representation rather than reproducing the original PCA dimensions. Huijgen et al. (2014) ~~and replicated the two-dimensional pattern for the Nazi-party scenario with Dutch students, but transfer of the item-rating approach to a slavery scenario produced weaker reliability and validity evidence~~. Huijgen et al. (2017) ~~, who confirmed the core theoretical structure while highlighting the difficulty of contextualization for most students~~. subsequently combined the closed item ratings with think-alouds and documented how students interpreted and sometimes misunderstood the response options. Thus, successful reuse of the Nazi scenario does not establish that the format transfers unchanged across topics or elicits the intended process. A recent Turkish qualitative case study of sixth- and eighth-grade students used the Hartmann and Hasselhorn framework and its components to analyze students' understanding of the Turkish Alphabet Reform (Aktın, Sarıççek, Karaçalı Taze, & Sağlam Tekir, 2026). The full-text report uses those components as deductive analytic categories; it does not establish direct administration or psychometric performance of this

nine-item instrument.

The difficulty of measuring HPT stems from its inherently interpretive character. Unlike factual recall, HPT cannot be ~~assessed through simple multiple-choice items with single correct answers~~ adequately inferred from decontextualized factual-recall items alone (Breakstone, 2014; Smith, 2017). Scenario-based instruments address this challenge by presenting contextual information and requiring students to evaluate statements reflecting different levels of historical reasoning (Hartmann & Hasselhorn, 2008; Seixas, Gibson, & Ercikan, 2015). However, this approach introduces a measurement tension: the richer the historical context, the more opportunity for construct-irrelevant factors—including attitudes toward that content—to influence responses.

2.2 Cross-Cultural Test Adaptation

The adaptation of educational and psychological instruments across languages and cultures requires systematic procedures to ensure that the adapted version measures the same construct as the original (Hambleton et al., 2005; International Test Commission, 2017). The International Test Commission (ITC) guidelines emphasize that adaptation involves more than translation: it requires attention to construct equivalence, cultural appropriateness, and technical quality of the adapted version (International Test Commission, 2017). For instruments used in history education, cross-cultural adaptation faces the additional challenge that historical content may carry different resonances across national contexts. A scenario about the Weimar Republic, originally designed for German students, intersects differently with Czech historical memory—while subsequent events

shape contemporary Czech memory of the interwar period, the scenario ~~centres~~centers on Weimar-era deliberation (early 1930s), which predates the 1938 Munich Agreement and cession of the Sudetenland by approximately ~~five~~eight years. Nevertheless, the broader period connects directly to the Nazi occupation of 1939–1945 in Czech collective memory.

In Czech lower-secondary education, history is prescribed within the compulsory People and Society curriculum area for Grades 6–9. The framework includes the 1920s and 1930s, fascism, Nazism, totalitarianism, and work with historical sources, but does not define HPT as a separate competence (Ministerstvo školství, mládeže a tělovýchovy, 2023, pp. 54–58, 140, 143). School curricula broadly align with this framework, although curriculum documents describe intended rather than enacted provision (Kaleta, 2025). Participating teachers reported that students had encountered the relevant interwar content; this does not establish equivalent prior instruction across schools.

This documentary account applies specifically to the lower-secondary subsample. The majority of participants were enrolled in upper-secondary education, for which this study did not conduct an equivalent curriculum- document audit. Eligibility at both levels required teacher confirmation that students had completed the interwar unit, but that confirmation is not evidence of uniform curricular coverage or instruction.

Cross-cultural validation studies should demonstrate that the adapted instrument pre-serves the factor structure, reliability, and validity relationships of the original (Hambleton et al., 2005). At minimum, this requires confirming the theorized factor structure through CFA, reporting reliability estimates appropriate for the item characteristics, and providing convergent validity evidence through expected correlations with related con-

structs. When the instrument will be used to compare groups or predict outcomes, measurement invariance testing becomes essential (American Educational Research Association et al., 2014).

2.3 Construct-Irrelevant Variance in Educational Assessment

The unified validity framework identifies construct-irrelevant variance as a principal threat to score interpretation (American Educational Research Association et al., 2014; Messick, 1995). CIV occurs when systematic variance in test scores is attributable to factors extraneous to the target construct. When assessment content intersects with respondents' attitudes or social identities, CIV can arise through attitude-consistent responding: students whose beliefs align with test content may find certain response options more natural or appealing, inflating their scores without corresponding gains in the assessed competence (Messick, 1995).

For the HPT instrument specifically, the concern is that the Weimar Republic scenario creates a *congruence pathway*—a mechanism whereby attitudinal similarity to item content lowers the cognitive effort required to endorse a response, inflating scores independently of competence. Consider a contextualization item such as “His decision was understandable given the political and economic turmoil of the Weimar Republic.” A student who endorses this statement because she has reconstructed the historical context produces a valid HPT response. But a student who endorses it because the sentiment resonates with her own political sympathies—not because she has engaged in historical reasoning—produces an inflated score. This would represent construct contamination in Messick’s

(1995) terms—a systematic source of score variance unrelated to HPT competence. Evidence of internal structure, including measurement invariance across relevant subgroups, constitutes essential validity evidence when scores are used to compare groups or when the construct definition requires demonstrating that extraneous factors do not influence scores (American Educational Research Association et al., 2014).

An alternative theoretical prediction deserves mention. Research on authoritarian cognition suggests that high authoritarianism is associated with reduced integrative complexity—the cognitive capacity to hold and reconcile multiple conflicting perspectives simultaneously (Duckitt, 2001; Suedfeld & Tetlock, 1977). Because HPT requires precisely this kind of integrative capacity, authoritarian students might perform *worse*, not better. The direction of any bias, if present, was genuinely uncertain. Additionally, dual-process theories of cognition (Evans & Stanovich, 2013) suggest that structured assessment formats may channel deliberative processing, overriding the automatic attitude-retrieval processes that would produce attitude-consistent responding. If so, the assessment format itself may buffer against attitudinal interference.

The construct boundary is therefore consequential. Under broader accounts of historical empathy, affective engagement and morally responsive evaluation can be target-relevant aspects of understanding a past actor (Barton & Levstik, 2004; Endacott & Brooks, 2013).

Under the narrower cognitive definition operationalized here, target variance consists of reconstructing context and integrating historically relevant considerations. Moral reaction or affect is not itself scored, and endorsement driven by option familiarity, task demands, or attitude congruence is construct-irrelevant when it substitutes for that reconstruction.

The rating data alone cannot determine which process produced an endorsement.

2.4 DIF and Measurement Invariance as Fairness Evidence

Differential item functioning and measurement invariance testing provide complementary approaches to evaluating measurement fairness (Chen, 2007; Cheung & Rensvold, 2002; Finch, 2005). DIF analysis examines whether individual items function differently across groups after controlling for the latent trait, identifying specific items that may be biased. Measurement invariance testing through multi-group CFA examines whether the overall measurement model holds across groups at three progressively restrictive levels: *configural invariance* (the same factors exist in both groups), *metric invariance* (items relate equally strongly to the factors in both groups), and *scalar invariance* (the scale origin is identical, so mean scores can be directly compared across groups).

For ordinal items, the weighted least squares mean and variance adjusted (WLSMV) estimator—a robust estimation method designed for categorical data—is recommended for CFA (Flora & Curran, 2004). Because ordinal items have discrete response categories rather than continuous scales, threshold-based invariance testing (which models the boundaries between response categories) replaces intercept-based approaches. The conventional criteria for evaluating invariance steps use changes in fit indices rather than chi-square difference tests, which are sensitive to sample size: $\Delta CFI \leq .01$ and $\Delta RMSEA \leq .015$ indicate that the more constrained model does not fit meaningfully worse (Chen, 2007; Cheung & Rensvold, 2002).

In the context of HPT measurement, establishing scalar invariance across ideology groups

would mean that the factor structure, loadings, and item thresholds are equivalent for students with different political attitudes—a necessary condition for concluding that observed score differences (or the lack thereof) reflect true construct differences rather than measurement artifacts.

2.5 The Present Study

This study addresses two interrelated measurement questions:

1. ~~Does~~ What psychometric evidence does the Czech adaptation of the Hartmann and Hasselhorn HPT instrument ~~demonstrate acceptable psychometric properties, including the theorized~~ provide, including evidence concerning a correlated three-factor ~~structure, adequate~~ representation, score reliability, and ~~convergent validity~~ convergence with historical knowledge?
2. ~~Does the instrument function fairly across students with different political attitudes, as evidenced by the absence of DIF and by scalar measurement invariance across ideology groups~~ What evidence do DIF, measurement invariance, and regression analyses provide about whether scores are affected by students' political attitudes?

~~We hypothesized that the three-factor structure would be confirmed and that the instrument would demonstrate measurement invariance across ideology groups. As convergent validity evidence, we expected historical knowledge to correlate positively with HPT scores, consistent with the theoretical conceptualization of HPT as a knowledge-dependent competence~~ The study and analysis plan were preregistered at <https://osf.io/zsngy>. The present measurement-focused organization, the primary

300 six-item HPT composite as the primary outcome, and several analyses were adopted
301 after registration, and not all registered analyses were implemented. Table S5 maps the
302 registered hypotheses, omissions, and post-registration additions; findings from the latter
303 are treated as exploratory or descriptive rather than as preregistered tests.

304 **3. Method**

305 **3.1 Adaptation Process**

306 The Czech HPT instrument was developed through a systematic translation and adap-
307 tation process following ITC guidelines (International Test Commission, 2017). Two
308 bilingual Czech-English experts conducted independent forward translations from the
309 ~~published English version of the instrument~~German instrument reported in Hartmann
310 (2008); wording was cross-checked against the published English validation article
311 (Hartmann & Hasselhorn, 2008). An independent backward translation was produced
312 by a third bilingual expert unfamiliar with the original. Five Czech history educators
313 at the secondary level reviewed the adapted items for conceptual equivalence, age-
314 appropriateness, and curricular alignment. No items required substantive modification;
315 minor wording adjustments were made to two items for naturalness in Czech. Reviewers
316 ~~confirmed that the Weimar Republic scenario content was familiar~~judged the scenario
317 content accessible to Czech secondary students ~~through the national history curriculum~~
318 ~~, given the direct connections between the interwar period and subsequent Czech~~
319 ~~historical experience (the Munich Agreement, the cession of the Sudetenland, and the~~
320 ~~Nazi occupation)~~given the curriculum coverage described in Section 2.2. This was

expert-review evidence, not direct evidence from student interviews.

The adaptation preserved the ~~original instrument's structure: the~~ four-point fit-to-situation response format (1 = ~~strongly disagree~~does not fit at all to 4 = ~~strongly agree~~fits very well), the ~~three-factor item structure~~three item categories (three items per ~~subscale~~category), and ~~the reverse-scoring of POP items were retained unchanged from the validated German version~~from the German source. All deviations from the original were documented transparently in the OSF repository (<https://doi.org/10.17605/OSF.IO/YNG37>). The ~~scenario narrative was~~author separately cross-checked the scenario against the Czech history curriculum~~to confirm that all contextual references were accessible to the target population, but no direct student response-process study was conducted.~~

No cognitive pilot study (e.g., think-aloud interviews) was conducted prior to main data collection, which constitutes a limitation of the adaptation process (see Section 5.6).

3.2 Participants

The sample comprised 293 students from 20 classrooms across 10 schools in the Czech Republic, spanning lower-secondary (International Standard Classification of Education [ISCED] level 2, approximately ages 11–15; $n = 87$, 29.7%) and upper-secondary (ISCED 3, approximately ages 15–19; $n = 206$, 70.3%) levels. The inclusion criterion was completion of the curricular unit on the interwar period. Schools were drawn from five Czech regions (Praha: 42.7%; Královéhradecký: 34.8%; Zlínský: 11.3%; Jihomoravský: 8.2%; Ústecký: 3.1%) and three funding categories (public: 90.8%; private: 6.5%; church-affiliated: 2.7%). Gender composition was 192 male (65.5%), 88 female (30.0%), and 13 identifying as an-

other gender (4.4%). Students' most recent history grade averaged 1.71 ($SD = 0.92$; Czech scale: 1 = excellent, 5 = failing).

Schools were recruited through convenience sampling; the sample over-represents upper-secondary students relative to the national age cohort composition and should not be treated as a probability sample of Czech adolescents. Sample size was determined by an a priori power analysis specified in the preregistered protocol (OSF: <https://doi.org/10.17605/OSF.IO/YNG37osf.io/zsngy>), targeting 80% power to detect standardized ideology effects of $\beta \geq .15$ in a two-level multilevel ~~model~~regression; the achieved $N = 285\text{--}293$ met ~~this target.~~that regression target. The calculation did not establish power for the post-registration DIF or MG-CFA analyses. Within each school, all participating students present on the day of administration completed the instrument battery during a regular class period (approximately 30–40 minutes). Teachers were present during administration but did not intervene in students' responses.

The sample's right-authoritarian attitude median was 2.33 on the five-point scale, and the historical-knowledge median was 3 of 6; fuller distributions appear in Section 4.4. No defensible student-, family-, or school-level socioeconomic indicator was collected. Consequently, the sample's socioeconomic composition and any associated selection bias cannot be characterized.

3.3 Instruments

3.3.1 Historical Perspective Taking (HPT) Students completed the Czech adaptation of the scenario-based HPT instrument from the German dissertation (Hartmann, 2008),

with the published English account used as supporting validation evidence (Hartmann & Hasselhorn, 2008). They read a narrative about a young man in the Weimar Republic considering joining an extremist political movement, then responded to nine items on a 4-point Likert scale in which Hannes discusses Germany's crisis and the 1930 election, then rated how well nine explanations fit his situation, including whether he could vote for an anti-democratic party such as the NSDAP, on a four-point fit-to-situation scale (1 = does not fit at all to 4 = fits very well). Three items each tapped presentism-present-oriented perspective (POP1–POP3), recognition of the agent's role role of the historical agent (ROA1–ROA3), and historical contextualization (CONT1–CONT3). POP items capture present-centered moral judgments of the historical figure (e.g., evaluating his decision by contemporary standards); these are reverse-scored (5 – POP, yielding a reversed range of 1–4) so that higher scores indicate more contextualized less present-oriented reasoning. ROA items reflect awareness of the agent's social position and circumstances without full contextual integration explain action through familiar or stereotyped social roles. CONT items represent the most sophisticated level, requiring require students to situate the historical figure's deliberation within the specific political, economic, and social conditions of the Weimar Republic. The primary composite was HPT_CTX6, six-item HPT composite was the mean of reversed POP and CONT items. HPT_TOT9 (adding ROA) served present-oriented perspective and contextualization scores and therefore represents the continuum between these two response categories. The nine-item total score (adding the theoretically less certain role-of-the-agent category) served only as a secondary sensitivity outcome.

3.3.2 Right-Authoritarian Ideology (FR-LF mini) A six-item short form of the Fragebogen Rechtsextremer Einstellungen—Leipziger Form (Questionnaire on Right-Wing Extremist Attitudes—Leipzig Form; Decker, Kiess, & Brähler, 2013), comprising acceptance of right-wing dictatorship (RD; 3 items) and National Socialist sympathy/relativization (NS; 3 items) facets, rated on a 5-point scale. The NS facet is theoretically the most direct operationalization of potential content congruence with the Weimar scenario.

3.3.3 Authoritarianism (KSA-3) The nine-item Kurzskala Autoritarismus (Short Scale of Authoritarianism; Beierlein, Asbrock, Kauff, & Schmidt, 2014), measuring authoritarian aggression, submission, and conventionalism on a 5-point scale. This instrument was used for a supplementary measurement invariance check using an alternative ideology operationalization.

3.3.4 Historical Knowledge (KN) Six multiple-choice items covering content relevant to the HPT scenario, scored dichotomously (sum score 0–6). This measure served as a convergent validity criterion, given the theoretical expectation that HPT is a knowledge-dependent competence.

3.3.5 Social Desirability (SDR-5) The five-item short form (Hays, Hayashi, & Stewart, 1989), rated on a 5-point scale with items SDR2–SDR4 reverse-coded. Included to evaluate whether HPT scores reflect impression management rather than genuine reasoning.

The source studies for the FR-LF included German respondents aged 14 years and older, whereas KSA-3 was developed for German-speaking adults aged 18 years and older and SDR-5 with adult medical outpatients and older adults. None therefore constitutes direct

Czech adolescent validation; suitability rests on expert review and the sample-specific evidence reported here and in Table S4.

3.4 Analytic Approach

All analyses were conducted in R (version ~~4.4.2~~4.6.1) using lavaan (0.7.2) for CFA, mirt (1.47) for IRT-based DIF analysis, psych (2.6.5) for reliability estimation, and lme4 ~~/performance-~~(2.0.6) for intraclass correlations.

Factor structure. CFA compared one-factor (general HPT), two-factor (POP+CONT vs. ROA), and three-factor (POP, ROA, CONT) models using the WLSMV estimator, appropriate for ordinal item responses (Flora & Curran, 2004). CFA fit indices summarize how well a hypothesized factor structure reproduces the observed correlations among items; multiple indices are reported because each captures a different aspect of misfit. Model evaluation used CFI ($\geq .95$ acceptable, $\geq .97$ good), the Tucker–Lewis Index (TLI; $\geq .95$), the Root Mean Square Error of Approximation (RMSEA; $\leq .08$ acceptable, $\leq .05$ good), and the Standardized Root Mean Square Residual (SRMR; $\leq .08$).

Reliability. Internal consistency was assessed ~~via using raw and polychoric~~ Cronbach's α ~~(raw and based on polychoric correlations) and~~ McDonald's ω , ~~which does not assume tau-equivalence (the assumption that all items relate equally strongly to the latent factor) and is generally preferred for congeneric measures (measures whose items may differ in how strongly they relate to the factor;)~~ and KR-20 for the dichotomous knowledge items. Reliability of the combined ideology score was estimated with the Mosier formula. Full methods and estimates are reported in Table S3.

Intraclass correlations. We computed intraclass correlation coefficients (ICCs) at the school and classroom levels to ~~characterise~~ characterize the multilevel structure of the data and to evaluate whether clustering warranted multilevel approaches.

Clustered data handling. The ~~DIF and~~ substantive regressions used random intercepts for the nested data. The DIF, MG-CFA analyses treat, and zero-order correlation analyses treated observations as independent, ~~which is standard practice in applied DIF and measurement invariance research. To evaluate whether ignoring the nested structure introduces meaningful bias, design effects (DEFF—a multiplier indicating how much clustering inflates standard errors relative to a simple random sample) were computed post hoc. A design effect of 1.0 means clustering has no impact on precision; values approaching 2.0 indicate modest inflation of standard errors that can usually be tolerated. The primary composite (HPT_CTX6) had DEFF = 1.56, within the conventional ≤ 2.0 threshold. The secondary composite (HPT_TOT9) had DEFF = 2.16, marginally above this threshold; consequently, inferential claims are anchored primarily to HPT_CTX6, and HPT_TOT9 results should be interpreted with this caveat.~~¹ Composite score ICCs do not establish that item thresholds, loadings, or their standard errors are unaffected by clustering; therefore, uncertainty for these analyses may be understated. The zero-order correlation intervals are reported as descriptive, unadjusted intervals rather than cluster-robust inference.

Missing data and score construction. Composite scores required at least two answered

¹ $DEFF = 1 + (\bar{n}_c - 1) \times ICC$, where $\bar{n}_c$ is the average cluster size (approximately 14.7 students per school). The TYPE=COMPLEX option in lavaan was not applied because the DIF analysis in mirt does not support clustered standard errors, and applying different corrections to DIF and MG-CFA would make the two sets of results less directly comparable.

items per three-item HPT subscale, four of six FR-LF items, seven of nine KSA-3 items, and four of five SDR-5 items; available responses were averaged. Missing knowledge responses were scored as incorrect, following the original analysis scripts. Reliability estimates used listwise-complete item sets, while correlations used pairwise-complete composite scores. These rules explain the analysis-specific sample sizes reported in the tables and supplement.

Differential item functioning. Low and High ideology groups were formed post-registration using tertile splits on a composite ideology score (mean of z-scored FR-LF and KSA-3), with the middle tertile excluded to maximize contrast ($\text{Low } n \approx = 96 \text{ per group}$). This extreme-group design follows established practice in DIF research. A constrained; High $n = 95$). A constrained unidimensional multi-group graded response model (GRM item discrimination and threshold-category-intercept parameters for each item by group are reported in Table S1 in the OSF-supplementary materials) using a sequential item-release design tested each item for uniform DIF (items are systematically easier or harder for one group at all ability levels) and non-uniform DIF (the group difference reverses depending on ability level) with Bonferroni correction (omnibus DIF by jointly freeing its discrimination and category intercepts, with Bonferroni-adjusted p -values ($\text{raw } p \times 9$) compared with a familywise $\alpha = .01$) of .05. The baseline model constrains all nine items to equality across groups; each test then frees the discrimination and threshold-parameters-category intercepts for the focal item, yielding a likelihood-ratio comparison against the fully constrained baseline (an all-others-as-anchor strategy). This approach assumes that the remaining eight items are DIF-free when testing each focal

item; iterative purification was not applied, which means that if multiple anchor items exhibited small DIF ~~in the same direction, anchor contamination could mask detection~~, focal-item detection could be affected. Because the correlated three-factor model was retained for theory- and continuity-based subscale reporting, the unidimensional GRM is a pragmatic sensitivity model rather than a fully aligned measurement model; multidimensionality or local dependence may affect the item tests.

Sensitivity of DIF detection. With approximately 96 respondents per ideology group, ~~the likelihood-ratio DIF tests at Bonferroni-corrected $\alpha = .01$ (9 tests) are powered primarily to detect moderate-to-large parameter differences ($\Delta\beta \geq 0.5$ logits on discrimination or threshold parameters). Weak DIF effects ($\Delta\beta < 0.5$ logits) cannot be confidently excluded. This sensitivity threshold is consistent with the sample size requirements documented for IRT-based DIF methods and nine Bonferroni-adjusted tests, statistical power was limited. No design-specific simulation was conducted, so the minimum detectable DIF magnitude is unknown.~~ The extreme-group design ~~maximizes statistical contrast between ideology subgroups increases ideological contrast~~ at the cost of ~~reduced per-group sample size—a standard trade-off in DIF research when the grouping variable is continuous~~ excluding the middle tertile and reducing group size.

Multi-group CFA. Stepwise MG-CFA tested configural, metric, and scalar invariance across ideology groups using the three-factor model with WLSMV estimation. Invariance was evaluated using $\Delta CFI \leq .01$ and $\Delta RMSEA \leq .015$ criteria (Chen, 2007; Cheung & Rensvold, 2002).

Substantive models. Random-intercept linear models predicted each standardized

HPT outcome from standardized FR-LF, KSA-3, historical-knowledge, and SDR-5 scores entered together, with intercepts for school and classroom. Thus, each reported ideology coefficient is mutually adjusted for the other ideology measure, knowledge, social desirability, and the nested grouping structure.

Convergent validity. Descriptive Pearson correlations between HPT composites and historical knowledge provided convergent validity evidence; Fisher-transformed intervals did not adjust for clustering.

Supplementary invariance check. To verify that invariance findings were not an artifact of the composite ideology definition, MG-CFA was also conducted using NS-only grouping (the facet most directly content-congruent with the Weimar scenario).

4. Results

The Results follow the two measurement questions in sequence. Sections 4.1–4.5 assess factor structure, score precision, distributions, and clustering; Sections 4.6–4.8 examine item and model-level fairness; Sections 4.9–4.10 then report the focal knowledge and ideology relationships.

4.1 Factor Structure

CFA supported the showed good fit for the correlated three-factor model (POP, ROA, CONT) as best-fitting, outperforming the two-factor and one-factor alternatives (Table 1). The three-factor model showed excellent global fit: Under the scaled WLSMV convention, this model had CFI = .999.975, TLI = .998.962, RMSEA = .011.039 (90% CI [.000, .051.067]),

and SRMR = .048. The two-factor model combining POP and CONT items on a single factor showed ~~acceptable but inferior~~ weaker fit (CFI = ~~.956~~.913, RMSEA = ~~.057~~.070), while the one-factor model fit poorly (CFI = ~~.926~~.866, RMSEA = ~~.073~~.086).

Table 1. CFA Fit Indices for HPT Factor Models (WLSMV Estimator)

Model	χ^2	df	CFI	TLI	RMSEA	90% CI
1-Factor	66.68 81.54	27	.926 .866	.902 .821	.073 .086	[.051 , .0
2-Factor (POP+CONT, ROA)	49.48 61.30	26	.956 .913	.940 .880	.057 .070	[.032 , .0
3-Factor (POP, ROA, CONT)	24.78 34.26	24	.999 .975	.998 .962	.011 .039	[.000, .0
Orthogonal bifactor (post-registration)	20.84	18	.993	.986	.024	[.000, .0

Note. WLSMV = weighted least squares mean and variance adjusted. All items were treated as ordered categorical. Reported chi-square, CFI, TLI, RMSEA, and RMSEA confidence-interval values use lavaan's scaled WLSMV convention; SRMR is the standard WLSMV value. The 2-factor model groups POP and CONT items on a single factor with ROA on a separate factor. ~~Chi-square values are mean and variance adjusted (WLSMV).~~ Nested model comparisons require the DIFFTEST procedure rather than standard chi-square difference tests. The bifactor model has a general factor and orthogonal three-item specific factors. Its comparison with the correlated three-factor model is non-nested and parameterization-dependent. All $p < .001$ except the three-factor model ($p =$ ~~.418~~.080) and bifactor model ($p =$.288).

The ~~near-perfect global fit of the three-factor model reflects the model's good global fit~~ should be interpreted alongside the small item set (~~nine items in three just identified~~

~~three-indicator factors—each subscale has exactly as many free parameters as observed~~
~~statistics, leaving no room for misfit within subscales) and should be interpreted~~
~~alongside local fit diagnostics: the largest modification index was 7.35 (POP1–ROA3~~
~~residual covariance; below the threshold of 10), and standardized residual correlations~~
~~were small ($|r| \leq .13$). With only and low-*df* specification: with three indicators per factor,~~
there is limited opportunity to reveal within-factor misfit. The Czech data are compatible
with the correlated three-factor specification, but this is not a test of cross-national
equivalence.

As a post-registration sensitivity analysis, an orthogonal bifactor model had more
favorable scaled fit indices (CFI = .993, RMSEA = .024, SRMR = .037; Δ CFI =
+.018 relative to the correlated three-factor model (9 items, 3 factors, 24 free parameters
~~against 45 observed statistics), there is minimal room for misfit; the near-perfect CFI~~
~~is therefore a structural consequence of the low-*df* specification rather than evidence~~
~~of an unusually well-fitting model. These results confirm that the Czech adaptation~~
~~preserves the theorized three-level structure of the original German instrument.). Thus,~~
fit indices favor the more complex bifactor model, rather than unequivocally favoring the
correlated three-factor model. That comparison is constrained by only three indicators per
specific factor, the alternative parameterization, and its non-nested status. We therefore
retain theory- and continuity-based subscale reporting as primary, while treating model
uncertainty as a limit on strong structural claims.

Inter-factor correlations from the three-factor CFA were: POP–ROA $\Phi = .377$, POP–
CONT $\Phi = .509$, ROA–CONT $\Phi = .631$. These moderate-to-strong correlations indicate

meaningful associations among the HPT dimensions, with ROA and CONT showing the strongest link. Discriminant validity is supported because all inter-factor correlations are below .85, confirming that the three subscales capture related but distinct aspects of historical perspective taking. are consistent with related but distinguishable item sets in this sample and provide bounded internal-structure evidence. They do not by themselves establish discriminant validity. Stronger evidence would require showing that the subscales relate differentially to external criteria in theoretically expected ways.

~~A supplementary bifactor model (one general factor plus three specific factors, in which each item loads on both the general factor and its subscale-specific factor) fit well ($\chi^2 = 14.70, df = 18, p = .682$) but did not meaningfully improve over the correlated three-factor model ($\Delta CFI = .001$). The low df (18) results from the additional cross-loadings consuming degrees of freedom, and the comparison with the correlated three-factor model ($df = 24$) should be treated cautiously given the different parameterisations. CONT items loaded primarily on the general factor (.47–.63), while POP and ROA retained meaningful specific factor variance, supporting subscale-level interpretation.~~

4.2 Standardized Factor Loadings

Table 2 presents the standardized loadings from the three-factor CFA. CONT items loaded most strongly (.61–.72), followed by POP items (.30–.74) and ROA items (.30–.67). POP2 showed the weakest loading (.30), at the conventional minimum threshold for item retention; it was retained to preserve fidelity to the original instrument structure. ~~The range of loadings across subscales reflects the heterogeneity within each three-item set and is~~

consistent with patterns reported in prior validations of short-form instruments.

Table 2. *Summary of Standardized Factor Loadings for the Three-Factor HPT Model*

Item-Subscale	Loading range
POP1 Present-oriented perspective	Presentism .30-.74
POP3 Role of historical agent	Presentism .30-.67
ROA3 Historical contextualization	Recognition of Agent's Role .67-.074 .53-.82
CONT3 Contextualization	CONT1 Contextualization .64-.054 .53-.74

Note. POP items are reverse-scored (higher = more contextualized reasoning). All loadings were significant at $p < .001$. WLSMV estimator with items treated as ordered categorical. Full item estimates, standard errors, and confidence intervals are reported in Table S2.

4.3 Reliability and Internal Consistency

Table 3 presents reliability coefficients for each HPT subscale. Raw summarizes reliability for the primary study scores. Across HPT subscales, raw Cronbach's α values ranged from .47 (POP) ranged from .46 to .64 (CONT); polychoric α (appropriate for ordinal items) ranged from .53 to .69; and McDonald's ω ranged from .55 to .69. CONT demonstrated the strongest reliability, consistent with its highest average factor loadings from .54 to .70. Full estimates for all HPT and predictor scores are in Table S3.

Table 3. *Reliability Coefficients Summary for HPT Subscales Primary Study Scores*

Subscale Score	Items/components	α (raw) Reliability	α (polychoric) ω total
POP Primary six-item HPT composite	3-6	.47-.66	.53-.55
ROA HPT total	3-9	.50-.70	.53-.56
CONT Combined ideology	3-2 scales	.64-.80	.69-.69

Note. POP items are reverse-scored. The Mosier estimate has a 95% bootstrap CI [.74, .84]. Full subscale and predictor alpha, polychoric-alpha, omega, and interval estimates are reported in Table S3.

Reliability was modest and constrains individual interpretation. FR-LF had α = Cronbach's alpha; ω .67, KSA-3 had α = McDonald's omegatotal. Mean r_{ii} .72, and the combined ideology score had estimated reliability .80, 95% CI [.74, .84]. Knowledge had KR-20 = mean inter-item correlation.

These values are modest by conventional standards but structurally constrained by the three-item design: with $k = 3$, even moderate inter-item correlations (.20-.35) yield .55, and SDR-5 was weak (α in this range. The Spearman-Brown formula (which estimates how reliability would improve if subscale length were doubled) projects α = .64-.79 for six-item subscales, though this assumes tau-equivalence—an assumption violated when loadings range from .30 to .74 (Table 2)—so the projections should be treated as approximate upper bounds. McDonald's ω (.55-.69) is more appropriate given unequal loadings (Table 2). The knowledge-HPT correlations reported below (r = .22-.34) demonstrate that the instrument carries genuine psychometric signal despite

modest subscale reliability. Correcting the ideology-HPT correlation for measurement unreliability (disattenuation—an adjustment that estimates what the correlation would be if both measures had perfect reliability) yielded a disattenuated $r = -.07$, indicating that measurement error does not mask a meaningful ideology association. However, reliability for the composite ideology score (the mean of z-scored FR-LF and KSA-3) was not separately estimated, which limits the precision of this disattenuated estimate.

4.4 Descriptive Statistics

Table 4 presents descriptive statistics. HPT scores followed the expected developmental ordering (POP_rev > ROA > CONT), with contextualization producing mean-score ordering: reversed present-oriented perspective scores were highest, followed by role-of-the-agent and contextualization scores. Contextualization produced the lowest scores ($M = 2.71$). Ideology scores were below the scale midpoint (this ordering may reflect item endorsement or difficulty and is not developmental evidence. Right-authoritarian attitude scores (FR-LF) and authoritarianism scores (KSA-3) were below their scale midpoints ($M = 2.48$; KSA: $M =$ and 2.86), reflecting the rarity of strong, respectively). The right-authoritarian endorsement in mainstream adolescent samples. attitude median was 2.33 (IQR [2.00, 3.00]) and 51 of 284 students (18.0%) scored above its midpoint. The authoritarianism median was 2.89 (IQR [2.53, 3.22]) and 106 of 283 students (37.5%) scored above its midpoint. On the mean of the two raw scales, 71 of 283 students (25.1%) scored above 3. Historical knowledge had a median of 3 (IQR [2, 4]), with 4.1% at the floor and 8.9% at the ceiling. No defensible student- or school-level socioeconomic measure was available.

Table 4. *Descriptive Statistics for Study Variables*

Variable	<i>n</i>	<i>M</i>	<i>SD</i>	Min	Max
HPT POP <u>Present-oriented perspective</u> (reversed)	287	2.97	0.64	1.33	4.00
HPT ROA <u>Role of the historical agent</u>	287	2.80	0.68	1.00	4.00
HPT CONT <u>Historical contextualization</u>	287	2.71	0.74	1.00	4.00
HPT CTX6 (primary) <u>Primary six-item HPT composite</u>	287	2.84	0.55	1.33	4.00
HPT TOT9 <u>Nine-item HPT total score</u>	287	2.83	0.49	1.67	3.89
KN Total <u>Historical knowledge</u> (0–6)	293	3.04	1.62	0.00	6.00
FR-LF RD <u>Dictatorship acceptance</u>	285	2.54	0.88	1.00	5.00
FR-LF NS <u>Nazi-crime relativization</u>	286	2.43	0.89	1.00	5.00
FR-LF Total <u>Right-authoritarian attitudes</u>	284	2.48	0.75	1.00	5.00
KSA-3 Total <u>Authoritarianism</u>	283	2.86	0.62	1.00	4.60
SDR-5 Total <u>Social desirability</u>	283	3.02	0.63	1.00	4.60

Note. HPT items scored on a 1–4 scale; POP items reverse-coded. FR-LF and KSA-3 scored on a 1–5 scale. KN scored as sum of correct responses (0–6).

4.5 Intraclass Correlations

~~Intraclass correlations indicated meaningful school-level clustering for most HPT outcomes (HPT_CTX6: ICC_school = .041; HPT_TOT9: ICC_school = .085; ROA: ICC_school = .102), consistent with between-school differences in instructional~~

emphasis and student composition. Classroom-level variance was near zero $/.000/.041$ for the primary composite but non-negligible for CONT (ICC_class = .033) and ROA (ICC_class = six-item HPT composite, .085/.000/.085 for the nine-item total score, .092/.011/.104). These ICCs justify multilevel modelling for substantive analyses. The school-level variance is consistent with between-school differences in history instruction, curricular emphasis, and student composition. The corresponding design effects for role of the historical agent, and .026/.007/.033 for contextualization. The zero classroom component for the primary composite were modest (HPT_CTX6: DEFF = 1.56; HPT_TOT9: DEFF = 2.16; see Section 3.4 for computation and implications). six-item HPT composite is consistent with the singularity warning in its focal model. These components describe score clustering but do not identify its source.

4.6 DIF Analysis

Table 5 presents omnibus DIF results for all nine HPT items. Using a constrained multi-group graded response model with likelihood-ratio tests and Bonferroni-adjusted significance ($\alpha = .01$), no items were flagged for either uniform or non-uniform DIF. All adjusted Bonferroni-adjusted p -values exceeded .01, indicating statistically equivalent item parameters across ideology groups. This means that at equal levels of underlying HPT competence, students with low and high ideological attitudes responded to each item in the same way—no individual item functioned differentially as a function of political attitudes were computed as raw $p \times 9$ and compared with a familywise α of .05 (equivalently, per-test $\alpha = .0056$). No DIF was detected between the low- and high-ideology groups with this sample and procedure; the magnitude that the design

could reliably detect is unknown.

Table 5. DIF Results per HPT Item (Likelihood-Ratio Tests from Multi-Group Graded Response Model)

Item	χ^2	df	p (raw)	p (Bonferroni)	Flagged
POP1	11.18 10.46	4	.025 .033	.222 .300	No
POP2	9.42 9.01	4	.051 .061	.462 .547	No
POP3	9.50 9.72	4	.050 .045	.448 .408	No
ROA1	9.88 9.59	4	.043 .048	.383 .432	No
ROA2	2.90 3.13	4	.575 .536	1.000	No
ROA3	8.03 8.25	4	.091 .083	.815 .744	No
CONT1	1.65 1.51	4	.800 .825	1.000	No
CONT2	3.45 3.60	4	.486 .462	1.000	No
CONT3	9.43 9.63	4	.051 .047	.461 .424	No

Note. Likelihood-ratio chi-square statistics from constrained multi-group graded response models. Each test compares a fully constrained baseline (all items equal across groups) to a model freeing discrimination and ~~threshold~~category-intercept parameters for the tested item. Low ($n = 96$) and High ($n = 9695$) ideology groups formed by tertile split on composite ideology score (middle tertile excluded). Bonferroni-adjusted p -values equal raw $p \times 9$ and were compared with familywise $\alpha = .01$ (~~9-tests~~.05 (equivalently, per-test $\alpha = .0056$)). 0 of 9 items were flagged. POP1 showed the largest DIF statistic ($\chi^2 = 11.1810.46$, $p_{\text{raw}} = .025.033$), well above the Bonferroni threshold after correction ($p_{\text{adj}} = .222.300$).

GRM item discrimination and ~~threshold~~ category-intercept parameters are reported in Table S1 in the OSF supplementary materials.

~~Although no individual item reached significance after Bonferroni correction, five of nine items (POP1, POP3, ROA1, ROA3, CONT3) had raw p -values below .10. This clustering of near-significant results is consistent with a pattern in which multiple items share weak DIF in the same direction—precisely the scenario under which the all-others-as-anchor strategy without iterative purification is most vulnerable to anchor contamination. Because the anchor set may itself be contaminated by small uniform DIF, the likelihood-ratio tests may underestimate the true magnitude of item-level differences. This pattern does not invalidate the overall null DIF finding, but it suggests that the absence of flagged items should~~ undetected DIF among those anchors could affect each test. The result should therefore be interpreted as “no moderate-to-large DIF detected” rather than DIF detected with this sample and procedure,” not “no DIF present.”

4.7 Measurement Invariance

Table 6 presents the stepwise MG-CFA results. The configural model ~~demonstrated adequate fit (produced~~ CFI = .978, .977, RMSEA = .043, .044, and SRMR = .074), ~~confirming that the three-factor structure holds separately in both ideology groups. Constraining factor loadings to equality across groups (metric invariance) produced no meaningful deterioration (.~~ Adding equality constraints did not worsen global fit: metric $\Delta\text{CFI} = .000, -.001$ and $\Delta\text{RMSEA} = -.002$). ~~Constraining both loadings and~~

thresholds (scalar invariance) similarly maintained fit ($\Delta\text{CFI} = +.002$, $\Delta\text{RMSEA} = -.006$). Both transitions satisfy the established criteria of $|\Delta\text{CFI}| \leq .01$ and $|\Delta\text{RMSEA}| \leq .015$. These changes fall within common fit-index criteria (Chen, 2007; Cheung & Rensvold, 2002), providing evidence for full scalar invariance. All three models converged and passed lavaan's post-estimation admissibility check; the diagonals of their model-implied residual-variance matrices were positive.

Table 6. Multi-Group CFA Fit Indices Across Invariance Levels (WLSMV)

Model	CFI	RMSEA	SRMR	ΔCFI	ΔRMSEA
Configural	.978 .977	.043 .044	.074	—	—
Metric	.978 .976	.041 .043	.081	.000 — .001	— .002 .001
Scalar	.980 .981	.035	.076	+ .002 .005	— .006 .008

Note. ΔCFI and ΔRMSEA computed relative to the preceding model. Invariance criteria: $|\Delta\text{CFI}| \leq .01$, $|\Delta\text{RMSEA}| \leq .015$ (Chen, 2007). Low and High ideology groups formed by tertile split on composite ideology ($n \approx 96$ per group); complete item-response data were available for 92 Low and 90 High participants. Items treated as ordered categorical.

The SRMR increased slightly from configural (.074) to metric (.081), marginally exceeding the conventional $\leq .08$ threshold. This threshold was developed for maximum-likelihood estimation with continuous indicators and may not apply directly to WLSMV-estimated models with polychoric residuals; the marginal exceedance should therefore be interpreted cautiously. The SRMR returned to .076 at the scalar step. This pattern can signal partial loading non-invariance. The largest ordinary modification index at

the metric step was 11.28 (a suggested cross-loading of 11.22 and suggested a POP2 cross-loading on the ROA factor in the Low ideology group), marginally above the conventional threshold of 10. This suggests a minor loading difference for POP2 that is consistent with its weak loading (.30) in the overall CFA (Table 2) rather than evidence of systematic group-specific misspecification group. Because ordinary modification indices ignore equality constraints, this is a local cross-loading diagnostic, not evidence of a between-group loading difference.

Negative estimated residual variances (Heywood cases) occurred for ROA items in the Low ideology group in the configural and metric models. Such estimates indicate a non-positive-definite covariance matrix, which can invalidate fit statistics and render parameter estimates uninterpretable. While Heywood cases can arise with small per-group samples and WLSMV estimation of ordinal items—particularly with three-item factors—they signal potential model inadmissibility. The admissible solutions and changes in CFI and RMSEA are compatible with scalar invariance under the fitted model. This remains qualified evidence rather than a benign estimation artefact. The per-group sample sizes ($n \approx 82-96$) are at the lower boundary of what MG-CFA with WLSMV requires. The configural and metric invariance conclusions should therefore be qualified: they represent the best available evidence given sample size constraints, but the Heywood cases introduce uncertainty about the precision of the configural baseline. The invariance results should be treated as convergent evidence alongside the DIF analysis rather than as definitive confirmation of scalar equivalence. firm equivalence conclusion: the analysis was added after registration, used only 90–92 complete cases per

extreme group, estimated three-indicator factors, and did not model school or classroom clustering.

4.8 Supplementary Invariance: NS-Only Grouping

To verify that invariance findings were not an artefact/artifact of the composite ideology grouping variable, we repeated the MG-CFA using groups defined by the NS facet alone (National Socialist sympathy/relativisation/relativization), which represents the most direct theoretical operationalisation/operationalization of content congruence with the Weimar scenario. With the NS tertile split, group sizes were $n = 126$ (Low) and $n = 125$ (High). The supplementary analysis largely supported scalar invariance, with 37 middle-tertile cases and 7 missing NS scores excluded. Fit did not deteriorate when equality constraints were added (configural CFI = .960.959, RMSEA = .050.051, SRMR = .070.071; metric Δ CFI = +.002, Δ SRMR = +.006.005; scalar Δ CFI = +.014.015, Δ SRMR = -.003). The scalar Δ CFI of .014 marginally exceeds the conventional $\leq .01$ threshold, indicating that full scalar invariance is not unambiguously confirmed for the NS-only grouping. The Because CFI increased and RMSEA and SRMR decreased at the scalar step, the positive Δ RMSEA remained within acceptable bounds, and the direction of CFI change was positive (improved fit), which is atypical for genuine invariance violations. Nevertheless, given that the NS facet is the most theoretically proximal grouping variable and the larger per-group sample sizes ($n = 126/125$) make the criterion breach more interpretable, partial scalar invariance—with possible threshold non-equivalence for individual items—cannot be ruled out. CFI does not indicate worse fit and should not be compared with a deterioration threshold by absolute magnitude. This post-registration

analysis provides descriptive robustness evidence, not a second confirmatory test. Full fit statistics for this supplementary analysis are available in the ~~OSF~~ analysis scripts in the ~~OSF~~ data and code repository.

4.9 Convergent Validity

Scenario-relevant historical knowledge correlated positively with all HPT outcomes: $r = .34$ (~~HPT_CTX6~~) for the primary six-item HPT composite, $r = .22$ (~~CONT~~), $.23$ for contextualization, and $r = .32$ (~~POP_rev~~). $.31$ for reversed present-oriented perspective. These moderate correlations indicate that students with stronger knowledge of the relevant historical period scored higher on HPT, ~~while also demonstrating that HPT is not reducible to knowledge alone—the shared variance (approximately 5–12%) leaves substantial HPT variance attributable to reasoning processes beyond factual recall. The scores were not empirically interchangeable, but their unshared variance cannot be attributed specifically to reasoning because it also includes measurement error and other unmeasured influences.~~ A limitation of this evidence is that the KN items were written to cover content directly relevant to the HPT scenario, so the correlation may partly reflect shared specific-content activation rather than the broader “knowledge-dependent competence” relationship posited theoretically. A general historical knowledge measure would provide a stronger convergent validity test.

History grade also correlated with ~~HPT_CTX6~~ the primary six-item HPT composite ($r = -.153$, $p = .010$; negative sign reflects the Czech scale where 1 = excellent). By contrast, ~~ideology measures showed negligible correlations with HPT (FR-LF-HPT_CTX6:~~

right-authoritarian attitudes showed a negligible correlation with the composite ($r =$
 $-.05$), and social desirability was unrelated to HPT outcomes ($r = -.02$ to $.01$).
 This pattern—significant knowledge correlations, null ideology and social desirability
 correlations—is consistent with an instrument that measures disciplinary reasoning
 rather than attitudes or impression management. One exploratory exception was a
 negative association between authoritarianism and reversed present-oriented perspective
 ($r = -.13$, 95% CI $[-.25, -.02]$). The corresponding mutually adjusted standardized
 KSA-3 coefficient was $\beta = -.166$, $SE = .066$, unadjusted $p = .013$. This observed
 direction was not predicted: registered H2 predicted the opposite direction on raw POP.
 No reported test implemented the registered Benjamini–Hochberg correction, so this
 association should be treated as hypothesis-generating rather than confirmed. Overall,
 the positive knowledge associations and absence of detected associations for the primary
 ideology–HPT relationship are consistent with, but do not establish, the intended score
 interpretation.

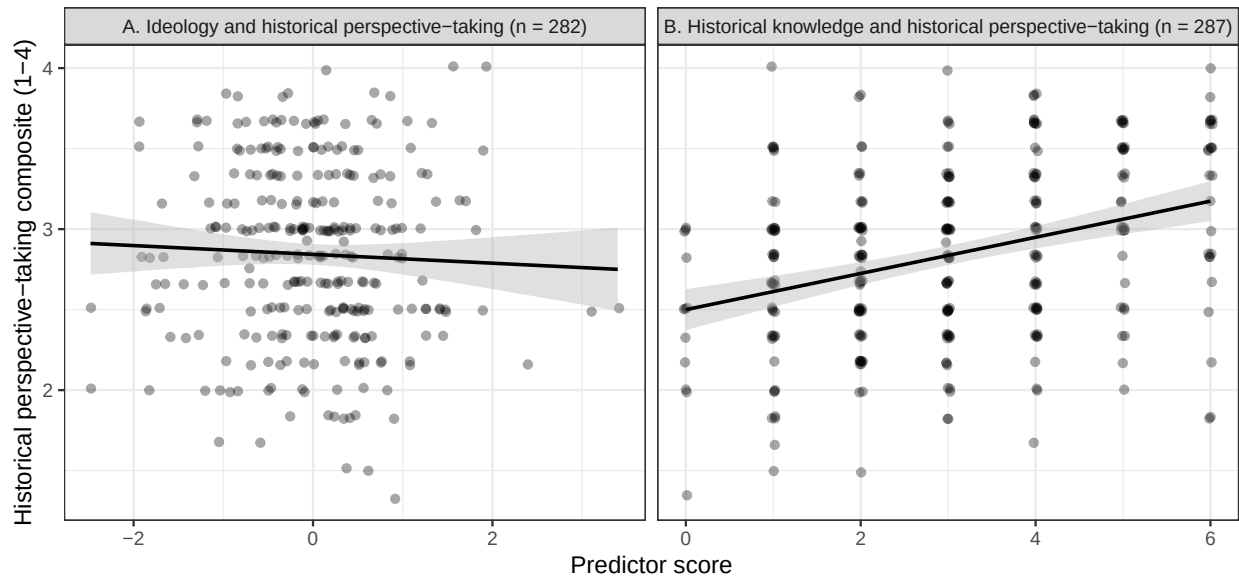
Table 7. Zero-Order Pearson Correlations Among Key Variables

	CTX6-HPT	CONT-Context	POP_rev- (reversed) Presentism	KN-Knowledge	FR-LF- attitudes Right-authoritarian	KSA-Authoritarianism
HPT-CTX6-Primary six-item HPT composite	—	[.78, .86]	[.71, .81]	[.23, .43]	[−.16, .07]	[−.15, .09]
HPT-CONT Contextualization	.82***	—	[.14, .36]	[.11, .33]	[−.10, .13]	[−.05, .19]

	<u>CTX6-HPT</u>	<u>CONT-Context</u>	<u>POP_rev- Presentism (reversed)</u>	<u>KN-Knowledge</u>	<u>FR-LF- Right-authoritarian attitudes</u>	<u>KSA-Authoritarianism</u>
<u>HPT</u>	.76***	.26***	—	[.20, .41]	[−.21, .02]	[−.25, −.02]
<u>POP_rev- Present-oriented perspective (rev.)</u>						
<u>KN-Total-Knowledge</u>	.34***	-.22.23***	-.32.31***	—	[−.22, .01]	[−.07, .16]
<u>FR-LF-Total- Right-authoritarian attitudes</u>	−.05	-.01-.02	−.09-.10	−.10	—	[.43, .60]
<u>KSA-3-Total- Authoritarianism</u>	−.03	.07	−.13*	.05	.52***	—
<u>SDR-5-Total-Social desirability</u>	−.02	.01	−.05	.03	−.19**	−.15

Note. Correlations are below the diagonal and Fisher-transformed, unadjusted descriptive
95% confidence intervals are above it. Pairwise complete observations ($n = 276-287$). HPT
POP-282-287; exact values in Table S3b). Present-oriented perspective is reverse-scored.
KN-Knowledge = historical knowledge (0–6 sum); FR-LF = right-authoritarian ideology
(1–5); KSA-3 = authoritarianism (1–5); SDR-5 = social desirability (1–5). Intervals and
p-values do not adjust for school or classroom clustering. *** $p < .001$. ** $p < .01$. * $p < .05$.
Figure 1 visualizes the near-zero ideology association and positive knowledge association
with the six-item HPT composite.

4.10 Substantive Context: Ideology-HPT Relationships



Points are jittered to show overlapping observations; lines and 95% confidence bands are unadjusted.

Figure 1: Observed relationships of the primary six-item HPT composite with the combined ideology score—the mean of standardized right-authoritarian attitude and authoritarianism scores—(Panel A) and historical knowledge (Panel B). Points are jittered to display overlap; lines and shaded regions show unadjusted linear fits and 95% confidence intervals.

~~Ideology was unrelated to HPT scores across all model specifications: FR-LF~~ The
 mutually adjusted standardized right-authoritarian attitude coefficient was near zero
 for the primary six-item HPT composite: $\beta = 0.012.016$, $SE = 0.067.068$, 95% CI $[-0.120,$
 $0.143.117, .149]$, $p = .863$ for the primary outcome (HPT_CTX6). ~~Historical~~ .817
 (analysis $n = 282$). The fitted model issued a singularity warning because the classroom
 random-intercept variance was estimated as zero; the school intercept and fixed effects
 remained estimable, and school fixed-effects sensitivity models gave the same substantive
 conclusion. Within the primary, preregistration-relevant model set, historical knowledge
 was the only consistent predictor ($\beta = 0.21-0.34$, $p < .001$). As an exploratory sensitivity
 analysis, equivalence tests used the Two One-Sided Tests (TOST) procedure, which
 reverses the logic of null-hypothesis testing by asking whether an effect is small enough

to be declared practically negligible. The Smallest Effect Size Of Interest (SESOI) was set post hoc at ± 0.20 ~~standardised~~ standardized units, corresponding to Cohen's (1988) threshold for a "small" effect. This SESOI is wider than the preregistered power target of $\beta \geq .15$, meaning that effects between .15 and .20 cannot be confidently excluded; the TOST results should therefore be treated as exploratory rather than confirmatory. With the ± 0.20 boundary, the TOST results indicated that ~~ideology~~ right-authoritarian attitude effects at or above $\beta = .20$ are statistically ruled out (~~all~~ for the primary relationship (TOST p ~~s~~ $\leftarrow .003 = .0034$; 90% CI ~~for the primary outcome~~ $[-0.10, 0.12, .096, .127]$, entirely within the ± 0.20 SESOI). The ~~ideology coefficient~~ right-authoritarian attitude coefficient for the primary six-item HPT composite remained non-significant across all sensitivity checks, including alternative HPT scoring composites, different ideology operationalizations, exclusion of high social desirability responders, random-slope models, and school fixed-effects specifications (~~all ps~~ $\rightarrow .43$). These substantive findings complement the measurement evidence: ~~the instrument not only measures equivalently across ideology groups (DIF, invariance) but also yields scores that are uncontaminated by political attitudes at the composite level~~ no DIF or composite-level ideology association was detected within the observed ideological range.

5. Discussion

5.1 Summary of Findings

The ~~HPT instrument passed every fairness test applied. Factor structure replicated, DIF was absent, scalar invariance held, and ideology~~ results provide qualified fairness

evidence. A correlated three-factor representation showed good fit, although the post-registration bifactor model had higher fit indices and model uncertainty limits strong structural claims. No DIF was detected with this procedure, and right-authoritarian attitudes did not predict HPT scores. This is the first psychometric evaluation of a Czech adaptation of the Hartmann and Hasselhorn HPT instrument and the first investigation of ideological fairness for any HPT measure. The primary six-item HPT composite; fit-index changes were compatible with scalar invariance, subject to small extreme groups and unmodeled clustering. The Czech adaptation demonstrated the theorized three-factor structure, acceptable reliability for three-item subscales, meaningful convergent validity with showed modest subscale reliability, a positive relationship with scenario-relevant historical knowledge, and sensitivity to school-level variation. Historical knowledge was the only construct that predicted HPT scores; political ideology, authoritarianism, and social desirability were all unrelated. An unadjusted exploratory authoritarianism-reversed-presentism association was not subjected to the preregistered multiplicity procedure; other ideology and social-desirability associations were not detected, although weak reliability constrains the latter null associations.

5.2 Factor-Structure Evidence

The correlated three-factor model (POP, ROA, CONT) fit substantially better than alternatives. The Czech data better than the registered one- and two-factor alternatives. However, the post-registration bifactor fit indices were higher, and the non-nested, confirming that the Czech adaptation preserves the dimensional structure of the original German instrument. This replication in a linguistically and educationally

distinct context confirms that the item grouping holds in a Czech sample, though with parameterization-dependent comparison means model uncertainty limits structural claims. This does not demonstrate cross-cultural equivalence, and the specification differs from the two-dimensional PCA pattern in the original German study. It shows only that the Czech data are compatible with separating the three item sets, under a near-saturated model the structural confirmation is relatively undemanding (see Section 4.1) with limited opportunity to reveal misfit. CONT items loaded most consistently (.61–.72), while POP2 and ROA2 showed weaker loadings (.30); these were retained for comparability but are candidates for revision in future adaptations.

5.3 Addressing Modest Reliability

The subscale reliabilities ($\alpha = .47.46-.64$; $\omega = .55.54-.69.70$) fall below conventional benchmarks but are structurally constrained by the three-item design—a feature of the original instrument, not a deficiency of the Czech adaptation. With k and limit individual-level interpretation. Three-item scales make high coefficients difficult to attain, but brevity does not make the observed precision adequate. Loevinger (1957) cautioned that highly homogeneous item sets can reduce a construct to repeated versions of a narrow question and argued for broad content representation; that principle supports balancing coverage and homogeneity, not treating low reliability as harmless. The six- and nine-item HPT scores were more precise ($\omega = 3-.66$ and inter-item correlations of .22–.37.70), Spearman-Brown projections yield $\alpha = .64-.79$ for six-item subscales. Despite modest reliability, the instrument detected theoretically expected knowledge-HPT relationships ($r = .22-.34$) and discriminated between HPT levels in the expected

ordering. The disattenuated ideology-HPT correlation ($r = -.065$) confirms that even with perfect reliability, no meaningful ideology association would emerge. We recommend using composite scores (HPT_CTX6 or HPT_TOT9) so we recommend composite scores rather than individual subscales for applied purposes. group-level research. Associations involving the least reliable measures, especially SDR-5, should be read cautiously because attenuation may obscure weak relationships.

5.4 Measurement Fairness Evidence

Scalar invariance was confirmed for the composite ideology grouping and largely supported for the NS-only analysis (with the caveat that the NS scalar ΔCFI marginally exceeded the conventional threshold; see Section 4.8). These results indicate that the HPT instrument imposes largely equivalent measurement demands on students regardless of their political attitudes. Fit index changes were compatible with scalar invariance for both ideology groupings, and the composite-group models were admissible in the current reproducible analysis. The small extreme groups, unmodeled clustering, and post-registration status nevertheless prevent a firm equivalence conclusion. The DIF analysis complements this at the item level: no item exhibited differential functioning after Bonferroni correction, which is particularly important for the CONT items most theoretically susceptible to ideological contamination, though the clustering of near-significant raw p -values warrants caution (see Section 4.6). adds item-level evidence, but no design-specific power analysis was conducted and the unidimensional GRM does not match the CFA structure. Together, these results provide qualified rather than definitive fairness evidence.

888 The correlation pattern—positive scenario-relevant knowledge–HPT associations ~~,-null~~
889 ~~ideology correlations,-~~(Chapman, 2021; Hartmann & Hasselhorn, 2008; Wineburg, 2001),
890 a near-zero primary ideology correlation, and null social desirability correlations—is con-
891 sistent with the intended score interpretation (American Educational Research Associa-
892 tion et al., 2014; Messick, 1995). However, the convergent evidence is bounded by the
893 scenario-proximal nature of the knowledge measure (see Section 4.9), which limits the
894 strength of the knowledge-dependent competence claim.

895 The near-zero ideology–HPT correlation ($r = -.05$) should not be interpreted as conclu-
896 sive evidence that ideology plays no role in HPT performance. As discussed in Section
897 2.3, two opposing pathways are theoretically plausible: a congruence pathway (whereby
898 attitudinal alignment inflates scores) and a reduced integrative complexity pathway
899 (whereby authoritarian cognition depresses ~~contextualisation~~contextualization). If both
900 pathways operate simultaneously, their effects could cancel, producing a near-zero net
901 correlation that masks two moderate-sized opposing effects. The present data cannot
902 distinguish a genuine null from cancelling pathways.

903 The present fairness evidence applies to mainstream Czech adolescent samples with re-
904 stricted ideological range. Users in contexts with greater political ~~polarisation~~polarization
905 should conduct their own DIF and invariance analyses before assuming equivalent mea-
906 surement.

5.5 Implications for Assessment Practice

The core practical message is that politically charged assessment content ~~does not~~ need not produce detectable construct-irrelevant ~~variance—but~~ variance in every application. In this study, no primary six-item HPT ~~this conclusion~~ composite association or DIF flag was detected at the available sensitivity. This pattern is bounded by the ~~present sample's~~ restricted ideological range and should not be generalized beyond the single structured, closed-ended ~~formats~~ scenario without further evidence.

The evaluation protocol used here ~~provides an evidence-based model~~ illustrates a possible workflow for evaluating fairness when assessment content intersects with respondents' attitudes. ~~This protocol is not specific to HPT or history education; it applies to any~~ Although developed for HPT, the workflow may inform studies of other scenario-based assessments where content carries attitudinal ~~valence—moral reasoning instruments~~ using politically divisive dilemmas, science literacy assessments touching evolution or climate change, and civic education assessments involving controversial policies valence.

The general approach involves five steps:

1. Identify the attitudinal dimension theoretically most likely to produce CIV.
2. Form extreme groups on that dimension.
3. Test DIF at the item level.
4. Test measurement invariance at the model level.
5. Examine ~~convergent and discriminant correlations to verify that construct scores~~ relate to theoretically expected variables and not to the attitudinal dimension

prespecified set of expected relationships, including whether HPT dimensions relate differentially to external outcomes as theory predicts.

The format itself may be a fairness mechanism. If scenario-based formats channel deliberative processing and override automatic attitude-retrieval (Evans & Stanovich, 2013), then the instrument format—not just the construct—buffers against motivated reasoning. However, this hypothesis has an important implication: if the format itself suppresses attitudinal responding, then the present fairness conclusion is conditional on the closed-ended ~~Likert~~ fit-to-situation rating format and cannot be extended to open-ended HPT assessments where ideology could operate through different cognitive pathways.

Several design features may explain the instrument’s resistance to attitudinal contamination: the scenario provides contextual information that channels deliberative processing, the ~~Likert~~ fit-to-situation items require evaluating propositions about a historical figure rather than expressing personal attitudes, and the school assessment context may activate an academic frame. These are hypotheses, not conclusions—the present study was not designed to test them experimentally, and doing so would require comparing structured and open-ended formats using the same content.

An alternative explanation for the ~~observed-measurement-equivalence-compatible~~ invariance fit pattern deserves consideration. In a school context where the scenario involves a Nazi-adjacent historical figure, students may uniformly avoid endorsing presentist responses—not because they are engaging in genuine historical reasoning, but because ~~contextualisation~~ contextualization is the socially safe response regardless of one’s actual political attitudes. If task-demand compliance produces equivalent response

patterns across ideology groups, the resulting ~~measurement~~ apparent invariance would reflect shared social desirability pressure rather than genuine absence of ideological influence on reasoning. The null SDR-5 correlations provide partial evidence against this interpretation, but a brief self-report measure of social desirability may not capture the specific task demands operating in a classroom assessment of politically sensitive historical content.

5.6 Limitations

Several limitations qualify these findings.

1. *Single country and scenario.* ~~The study uses one national context and one historical scenario; whether the fairness findings generalise to other contexts remains to be established.~~ Nine ratings of one Weimar scenario do not provide nine independent samples of the HPT content domain, so scenario-specific wording, knowledge, or moral salience cannot be separated from HPT. Future studies should use multiple scenarios spanning different periods, places, and political contexts.
2. *Restricted ideological range.* ~~Only 17.7%~~ Depending on the measure, 18.0% (FR-LF), 37.5% (KSA-3), or 25.1% (their raw-score mean) of respondents scored above the scale midpoint ~~on ideology measures~~, so the DIF and invariance analyses compare moderately elevated to low ideology, not genuine political extremism. The null ideology finding should be interpreted as “no evidence of CIV in this range-restricted sample” rather than as a general demonstration that politically charged content is fair.

3. *Limited DIF power.* With $n \approx 96$ per group, ~~the analysis was powered primarily to detect moderate-to-large effects ($\Delta\beta \geq 0.5$ logits); weak DIF effects cannot be confidently excluded~~DIF power was limited. A formal power simulation for the GRM likelihood-ratio test at Bonferroni-corrected α was not conducted; ~~future work should quantify, so~~ the minimum detectable DIF magnitude ~~for this design is~~unknown.
4. *Gender imbalance.* The sample was 65.5% male. ~~Gender is a documented moderator of authoritarian attitude endorsement in adolescent samples,~~ and no DIF or sensitivity analysis by gender was conducted. ~~If male students drive both ideology variance and the null finding, the fairness conclusion may not hold for~~The fairness conclusion should therefore be replicated in a more gender-balanced samples.
5. Convenience sampling and unavailable socioeconomic data. The sample over-represented upper-secondary students and was not probabilistic. Because no defensible socioeconomic indicator was collected, neither socioeconomic coverage nor selection along that dimension can be assessed.
6. *Modest reliability.* Subscale reliabilities ($\alpha = .47$.46–.64; $\omega = .54$.70) constrain individual-level score interpretations; future adaptations should consider expanding subscale length. ~~Omega hierarchical (ω_h) from the bifactor model was not computed; this index would clarify what proportion of subscale variance is attributable to the general HPT factor versus specific subscale factors.~~
7. *Method and format constraints.* All measures were self-report instruments administered in a single session. The closed-ended ~~Likert~~fit-to-situation rating format may

not ~~generalise~~ generalize to open-ended HPT assessments where ideology could operate through different cognitive mechanisms (Smith, 2017; Smith, Breakstone, & Wineburg, 2018).

8. *No response process evidence.* ~~The critical unanswered question is whether Czech students actually invoke historical reasoning when responding or whether the structured format produces contextualising responses through a different cognitive pathway. Cognitive interview studies could resolve this.~~ Think-aloud or cognitive- interview studies, extending Huijgen et al. (2017), could compare the same content in rating and open-response formats. Evidence use and integration of conflicting historical considerations would support contextual reconstruction; explicit moral evaluation or affective engagement would identify processes outside this instrument's narrow cognitive scoring target; paraphrase and option- matching comments would indicate format-driven responding; and endorsement tied to one's present political position would indicate attitude congruence. These observable indicators could distinguish pathways that the present ratings cannot separate.

5.7 Recommendations

For researchers developing or adapting HPT instruments: (a) report ω alongside α for short subscales, with Spearman-Brown projections for context; (b) evaluate fairness through DIF and measurement invariance for theoretically relevant grouping variables; (c) consider revising weak-loading items (POP2, ROA2) while preserving the theoretical structure; (d) expand subscale length to improve reliability; and (e) replicate fairness

evaluations across diverse samples, scenarios, and cultural contexts.

5.8 Conclusion

The Czech ~~adaptation~~ data were compatible with a correlated three-factor representation of the Hartmann and Hasselhorn HPT ~~instrument preserves the three-factor structure of~~ the original, produces item sets and produced scores that relate to scenario-relevant historical knowledge in theoretically expected ways, ~~and functions with largely equivalent measurement properties across students with different political attitudes.~~ Fit-index patterns were compatible with measurement invariance across ideology groups, but small extreme groups, unmodeled clustering, and post-registration analysis prevent a firm equivalence conclusion. At the sensitivity levels afforded by the present sample and in the observed ideological range, ~~political ideology was unrelated to HPT scores across all model specifications.~~ right-authoritarian attitudes were unrelated to the primary six-item HPT composite. An unadjusted exploratory authoritarianism association with one subscale was not subjected to the preregistered multiplicity procedure.

For researchers and assessment developers, the practical implication is that structured scenario-based assessments can ~~employ politically sensitive historical content without detectable measurement bias—at least in mainstream classroom settings with restricted ideological range~~ be evaluated for attitude-related measurement differences using complementary item-, model-, and relationship-level checks. In this sample, those checks detected no bias in the primary six-item HPT composite at the available sensitivity. Whether this ~~conclusion~~ pattern extends to more ~~polarised~~ polarized populations, open-

ended assessment formats, or different historical scenarios remains an open empirical question. The evaluation protocol ~~demonstrated~~ illustrated here—CFA, DIF, MG-CFA, and ~~convergent/discriminant correlations—provides a replicable~~ a prespecified network of external relationships—provides a possible template for answering that question wherever attitudinally charged content meets competence assessment.

Data Availability Statement

De-identified data, annotated analysis scripts, the complete instrument battery (in Czech), ~~supplementary materials~~, and all deviations from the original instrument are available ~~on the Open Science Framework in the OSF data and materials repository~~ (<https://doi.org/10.17605/OSF.IO/YNG37>). Data are provided in machine-readable formats (RDS and xlsx) accompanied by a full variable codebook (PDF). The ~~analysis pipeline consists of seven numbered R Markdown scripts that reproduce all reported results~~ pipeline contains five numbered analytic R Markdown reports, two documentation reports, shared scoring helpers, and a revision-reporting script; a README ~~file~~ documents the directory structure, execution order, software requirements, and mapping of scripts to manuscript tables and figures. Supplementary materials (~~Table S1: GRM item parameters~~) Tables S1–S5 accompany this manuscript and are included in the ~~OSF repository~~ package. The package is currently available at OSF and is prepared for a planned archival migration to Zenodo; it has not yet been migrated. The study design, hypotheses, and analysis plan were preregistered on OSF (<https://osf.io/zsngy>) prior to data collection; deviations and omissions are disclosed in Table S5.

Conflict of Interest Disclosure

The author declares that he complies with the PCI rule of having no financial conflicts of interest in relation to the content of the article, and has no non-financial conflicts of interest.

Ethics Statement

This study involved secondary school students completing questionnaires during regular class periods. Under Czech law (Act No. 110/2019 Sb., on the processing of personal data), ethics committee approval is not required for non-interventional educational research involving anonymous questionnaires where no personal data are collected. Participation was voluntary; students were informed about the study's purpose and their right to decline participation without consequences. All responses were de-identified at the point of data entry. No personally identifiable information was collected or stored. The study followed the principles of the Declaration of Helsinki for research involving human participants.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Author Contributions (CRediT)

Juda Kaleta: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization, Project administration.

Acknowledgements

The author thanks the participating schools, teachers, and students for their cooperation. The translation panel members and expert reviewers who contributed to the adaptation process are acknowledged with gratitude.

AI Use Disclosure

The following AI tools were used during the preparation of this manuscript:

- *Brainstorming and scripting assistance*: Claude (Anthropic) — used for exploring conceptual framing, structural alternatives, and scripting support for data processing and analysis workflows.
- *Language editing*: Grammarly — used for grammar and style checking.

All research design decisions, analytical choices, theoretical interpretations, and the final text are solely the author's responsibility. The author reviewed, revised, and approved all content.

References

- Aktın, K., Sariçiçek, A., Karaçalı Taze, H., & Sağlam Tekir, H. (2026). Taking a historical perspective: Students' understanding of the Turkish Alphabet Reform (November 1928). *Innovation: The European Journal of Social Science Research*, 1-35. Retrieved 2026-10-02, from <https://doi.org/10.1080/13511610.2026.2630018> doi: 10.1080/13511610.2026.2630018
- Alvén, F. (2019). Bias in teachers' assessments of students' historical narratives. *History Education Research Journal*, 16(2), 306-321. doi: 10.18546/HERJ.16.2.10
- American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (2014). *Standards for Educational and Psychological Testing*. American Educational Research Association. Retrieved 2025-02-15, from <https://www.testingstandards.net/open-access-files.html> (Prepared by the Joint Committee on Standards for Educational and Psychological Testing of the American Educational Research Association, American Psychological Association, and National Council on Measurement in Education.)
- Badham, L., Meadows, M., & Baird, J.-A. (2025). Construct comparability and the limits of post hoc modeling: insights from International Baccalaureate multi-language assessments. *Frontiers in Education*, 10, 1-20. doi: 10.3389/feduc.2025.1616879
- Barton, K. C., & Levstik, L. S. (2004). *Teaching history for the common good*. Lawrence Erlbaum Associates. doi: 10.4324/9781410610508
- Beierlein, C., Asbrock, F., Kauff, M., & Schmidt, P. (2014). *Die Kurzskala Autoritarismus (KSA-3): Ein ökonomisches Messinstrument zur Erfassung autoritärer Einstellungen* (No.

- 1113 2014/32). GESIS - Leibniz-Institut für Sozialwissenschaften.
- 1114 Breakstone, J. (2014). Try, Try, Try Again: The Process of Designing New History As-
 1115 sessments. *Theory & Research in Social Education*, 42(4), 453-485. Retrieved 2025-
 1116 05-06, from <https://doi.org/10.1080/00933104.2014.965860> doi: 10.1080/
 1117 00933104.2014.965860
- 1118 Chapman, A. (2021). Introduction: Historical knowing and the 'knowledge turn'. In
 1119 A. Chapman (Ed.), *Knowing History in Schools: Powerful Knowledge and the Powers*
 1120 *of Knowledge* (p. 1-31). UCL Press. Retrieved 2025-03-27, from [https://doi.org/](https://doi.org/10.14324/111.9781787357303)
 1121 [10.14324/111.9781787357303](https://doi.org/10.14324/111.9781787357303) doi: 10.14324/111.9781787357303
- 1122 Chen, F. F. (2007). Sensitivity of goodness of fit indexes to lack of measurement invari-
 1123 ance. *Structural Equation Modeling: A Multidisciplinary Journal*, 14(3), 464–504. doi:
 1124 10.1080/10705510701301834
- 1125 Cheung, G. W., & Rensvold, R. B. (2002). Evaluating goodness-of-fit indexes for testing
 1126 measurement invariance. *Structural Equation Modeling: A Multidisciplinary Journal*,
 1127 9(2), 233–255. doi: 10.1207/S15328007SEM0902_5
- 1128 *Children's concepts of empathy and understanding in history.* (1987).
- 1129 Činátl, K., Najbert, J., & Ripka, V. (2021). Historická gramotnost v aplikaci HistoryLab - re-
 1130 alistický přístup k osvojování didaktické teorie dějepisu. *Historie – Otázky – Problémy*,
 1131 13(2), 50-70. Retrieved 2022-07-12, from [http://hdl.handle.net/20.500.11956/](http://hdl.handle.net/20.500.11956/170260)
 1132 [170260](http://hdl.handle.net/20.500.11956/170260)
- 1133 Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence
 1134 Erlbaum Associates.
- 1135 Decker, O., Kiess, J., & Brähler, E. (2013). *Rechtsextremismus der Mitte: Eine sozialpsycholo-*

- 1136 gische Gegenwartsdiagnose. Psychosozial-Verlag. doi: 10.30820/9783837966367
- 1137 Duckitt, J. (2001). A dual-process cognitive-motivational theory of ideology and prejudice.
- 1138 In *Advances in experimental social psychology* (Vol. 33, pp. 41–113). Elsevier. doi: 10
- 1139 .1016/S0065-2601(01)80004-6
- 1140 Endacott, J. L. (2014). Negotiating the Process of Historical Empathy. *Theory &*
- 1141 *Research in Social Education, 42*, 4-34. Retrieved from [https://doi.org/10.1080/](https://doi.org/10.1080/00933104.2013.826158)
- 1142 00933104.2013.826158 doi: 10.1080/00933104.2013.826158
- 1143 Endacott, J. L., & Brooks, S. (2013). An Updated Theoretical and Practical Model for
- 1144 Promoting Historical Empathy. *Social Studies Research and Practice, 8*, 41-58. doi:
- 1145 10.1108/ssrp-01-2013-b0003
- 1146 Evans, J. S. B. T., & Stanovich, K. E. (2013). Dual-process theories of higher cognition:
- 1147 Advancing the debate. *Perspectives on Psychological Science, 8*(3), 223–241. doi: 10
- 1148 .1177/1745691612460685
- 1149 Finch, H. (2005). The MIMIC model as a method for detecting DIF: Comparison with
- 1150 Mantel-Haenszel, SIBTEST, and the IRT likelihood ratio. *Applied Psychological Mea-*
- 1151 *surement, 29*(4), 278–295. doi: 10.1177/0146621605275728
- 1152 Flora, D. B., & Curran, P. J. (2004). An empirical evaluation of alternative methods of
- 1153 estimation for confirmatory factor analysis with ordinal data. *Psychological Methods,*
- 1154 9(4), 466–491. doi: 10.1037/1082-989X.9.4.466
- 1155 Hambleton, R. K., Merenda, P. F., & Spielberger, C. D. (Eds.). (2005). *Adapting educational*
- 1156 *and psychological tests for cross-cultural assessment*. Lawrence Erlbaum Associates. doi:
- 1157 10.4324/9781410611758
- 1158 Hartmann, U. (2008). *Perspektivenübernahme als eine Kompetenz historischen Verstehens* (Doc-

- toral dissertation, Georg-August-Universität Göttingen). doi: 10.53846/goediss-290
- Hartmann, U., & Hasselhorn, M. (2008). Historical perspective taking: A standardized measure for an aspect of students' historical thinking. *Learning and Individual Differences, 18*, 264-270. Retrieved 2025-05-06, from <http://dx.doi.org/10.1016/j.lindif.2007.10.002> doi: 10.1016/j.lindif.2007.10.002
- Hays, R. D., Hayashi, T., & Stewart, A. L. (1989). A Five-Item Measure of Socially Desirable Response Set. *Educational and Psychological Measurement, 49*, 629-636. Retrieved from <https://doi.org/10.1177/001316448904900315> doi: 10.1177/001316448904900315
- Huijgen, T., van Boxtel, C., van de Grift, W., & Holthuis, P. (2014). Testing elementary and secondary school students' ability to perform historical perspective taking: the constructing of valid and reliable measure instruments. *European Journal of Psychology of Education, 29*, 653-672. Retrieved from <https://doi.org/10.1007/s10212-014-0219-4> doi: 10.1007/s10212-014-0219-4
- Huijgen, T., van Boxtel, C., van de Grift, W., & Holthuis, P. (2017). Toward Historical Perspective Taking: Students' Reasoning When Contextualizing the Actions of People in the Past. *Theory & Research in Social Education, 45*, 110-144. Retrieved from <https://doi.org/10.1080/00933104.2016.1208597> doi: 10.1080/00933104.2016.1208597
- International Test Commission. (2017). *ITC guidelines for translating and adapting tests (second edition)*. International Test Commission. Retrieved from https://www.intestcom.org/files/guideline_test_adaptation_2ed.pdf
- Jost, J. T., Glaser, J., Kruglanski, A. W., & Sulloway, F. J. (2003). Political conservatism as motivated social cognition. *Psychological Bulletin, 129*(3), 339-375. doi: 10.1037/

0033-2909.129.3.339

- Kahan, D. M., Peters, E., Dawson, E. C., & Slovic, P. (2017). Motivated numeracy and enlightened self-government. *Behavioural Public Policy*, 1(1), 54–86. doi: 10.1017/bpp.2016.2
- Kaleta, J. (2025). Sticking to tradition: history content distribution in Czech school curricula. *History Education Research Journal*, 22. Retrieved 2025-07-07, from <https://doi.org/10.14324/herj.22.1.14> doi: 10.14324/herj.22.1.14
- Lee, P., & Ashby, R. (2001). Empathy, perspective taking, and rational understanding. In O. L. Davis, E. A. Yeager, & S. Foster (Eds.), *Historical empathy and perspective taking in the social studies* (p. 21-50). Rowman & Littlefield.
- Lee, P., & Shemilt, D. (2003). A scaffold, not a cage: Progress and progression models in history. *Teaching History*, 113, 13–23.
- Loevinger, J. (1957). Objective Tests as Instruments of Psychological Theory. *Psychological Reports*, 3(3), 635-694. doi: 10.2466/pr0.1957.3.3.635
- Maggioni, L., VanSledright, B., & Alexander, P. A. (2009). Walking on the Borders: A Measure of Epistemic Cognition in History. *The Journal of Experimental Education*, 77(3), 187-214. doi: 10.3200/jexe.77.3.187-214
- Messick, S. (1995). Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning. *American Psychologist*, 50, 741-749. doi: 10.1037/0003-066x.50.9.741
- Ministerstvo školství, mládeže a tělovýchovy. (2023). *Rámcový vzdělávací program pro základní vzdělávání*. MŠMT. Retrieved 2024-12-17, from https://www.edu.cz/wp-content/uploads/2023/07/RVP_ZV_2023_cista_verze.pdf

- 1205 Ripka, V., Sýkorová, P., Münich, J., & Chvojka, E. (2024). Epistemic Cognition Triangu-
 1206 lated: What Can We Learn about the Theory of Epistemic Beliefs in History from
 1207 Reassessment of Its Measurement. In H. Å. Elmersjö & P. Zanazanian (Eds.), *Teach-*
 1208 *ers and the Epistemology of History* (p. 307-324). Springer Nature Switzerland. Re-
 1209 trieved 2025-09-24, from https://doi.org/10.1007/978-3-031-58056-7_17 doi:
 1210 10.1007/978-3-031-58056-7_17
- 1211 Seixas, P., Gibson, L., & Ercikan, K. (2015). A Design Process for Assessing Historical
 1212 Thinking: The Case of a One-Hour Test. In *New Directions in Assessing Historical*
 1213 *Thinking* (p. 102-116). Taylor & Francis.
- 1214 Seixas, P., & Morton, T. (2013). *The Big Six - Historical Thinking Concepts*. Nel-
 1215 son Education. Retrieved 2025-01-01, from [https://archive.org/embed/](https://archive.org/embed/bigsixhistorical0000seix)
 1216 [bigsixhistorical0000seix](https://archive.org/embed/bigsixhistorical0000seix)
- 1217 Seixas, P., & Peck, C. (2004). Teaching historical thinking. In A. Sears & I. Wright (Eds.),
 1218 *Challenges and Prospects for Canadian Social Studies* (pp. 109–117). Vancouver, BC:
 1219 Pacific Education.
- 1220 Shemilt, D. (1984). Beauty and the philosopher: Empathy in history and classroom. In
 1221 A. K. Dickinson, P. J. Lee, & P. J. Rogers (Eds.), *Learning history* (pp. 39–84). Heine-
 1222 mann Educational Books.
- 1223 Smith, M. (2017). Cognitive Validity: Can Multiple-Choice Items Tap Historical Thinking
 1224 Processes? *American Educational Research Journal*, 54(10), 1256-1287. doi: 10.3102/
 1225 0002831217717949
- 1226 Smith, M., Breakstone, J., & Wineburg, S. (2018). History Assessments of Thinking: A
 1227 Validity Study. *Cognition and Instruction*, 37(1), 118-144. doi: 10.1080/07370008.2018

- 1228 .1499646
- 1229 Suedfeld, P., & Tetlock, P. (1977). Integrative complexity of communications in in-
 1230 ternational crises. *Journal of Conflict Resolution*, 21(1), 169–184. doi: 10.1177/
 1231 002200277702100108
- 1232 van Boxtel, C., & van Drie, J. (2018). Historical Reasoning. In S. Metzger & L. Harris
 1233 (Eds.), *The Wiley International Handbook of History Teaching and Learning* (p. 149-176).
 1234 Wiley. doi: 10.1002/9781119100812.ch6
- 1235 van Drie, J., & van Boxtel, C. (2008). Historical Reasoning: Towards a Framework for
 1236 Analyzing Students' Reasoning about the Past. *Educational Psychology Review*, 20,
 1237 87-110. Retrieved from <http://dx.doi.org/10.1007/s10648-007-9056-1> doi:
 1238 10.1007/s10648-007-9056-1
- 1239 VanSledright, B. (2014). *Assessing historical thinking and understanding: Innovative designs*
 1240 *for new standards*. Routledge. doi: 10.4324/9780203464632
- 1241 Wineburg, S. (2001). *Historical Thinking and Other Unnatural Acts - Charting the Future of*
 1242 *Teaching the Past*. Temple University Press.

TOP Checklist and Disclosures Form

This checklist must be completed and included as an appendix at the end of your preprint prior to submission to PCI Psychology. Manuscripts without this document included as an appendix will be returned to authors without review.

The policy of PCI Psychology is to recommend papers only if the data, methods used in the analysis, and any digital materials used to conduct the research are clearly and precisely documented and are maximally available to any researcher for purposes of reproducing the results or replicating the procedure. PCI Psych follows the principle of “as open as possible, as closed as necessary.”

First author name (last/family, first/given): Kaleta, Juda

Preprint DOI or URL: https://doi.org/10.31234/osf.io/hxngm_v2

Section 1: Data

Does your manuscript contain reports of any data?

☒ **Yes** (continue with next question)

☐ **No** (skip to Section 2)

Are appropriately ~~anonymised~~ anonymized raw data available within a trusted digital repository?

☒ **Yes, available at this link:** <https://doi.org/10.17605/OSF.IO/YNG37>

1261 ☐ No, justification:

1262 **Are third-party data cited in the manuscript, with a DOI?** (e.g., for preexisting data, data
1263 deposited in a repository)

1264 ☐ Yes, the DOI is as follows:

1265 **[X] No, justification:** The data were collected specifically for this study; no third-party
1266 datasets were used.

1267 **Is there a data dictionary and/or readme file included with the data to make it inter-**
1268 **pretable?**

1269 **[X] Yes, available at this link:** <https://doi.org/10.17605/OSF.IO/YNG37> (full variable
1270 codebook in PDF and README file)

1271 ☐ No, justification:

1272 **Do you indicate in the manuscript how the sample size was determined?**

1273 **[X] Yes.** (Section [3.43.2](#): a priori power analysis targeting 80% power to detect $\beta \geq .15$)

1274 ☐ No, justification:

1275 **Do you report all data exclusions (e.g., outliers, careless responders)?**

1276 **[X] Yes.** (Section 3.4 and [4.6: Table S5: the middle ideology tertile excluded for DIF](#)
1277 [extreme-group design; all exclusions reported](#)[was excluded for post-registration DIF and](#)
1278 [MG-CFA analyses; this analysis-specific exclusion is reported.](#))

1279 ☐ No, justification:

1280 **Do you report all inclusion/exclusion criteria and when they were established?**

1281 **[X] Yes.** (~~Inclusion/exclusion criteria~~ Participant eligibility is described in Section 3.2
1282 and was preregistered; post-registration analysis-specific exclusions are described in Sec-
1283 tions ~~3.4 –3.5 and were preregistered prior to data collection~~ and Table S5.)

1284 ☐ No, justification:

1285 **Are all measures, questions, and/or conditions used in the study described in the**
1286 **manuscript or available in the supplemental material?**

1287 **[X] Yes.** (All instruments described in Sections 3.1–3.3; full Czech-language instrument
1288 battery available on OSF: <https://doi.org/10.17605/OSF.IO/YNG37>)

1289 ☐ No, justification:

1290 **Section 2: Analysis Scripts/Code/Codebooks**

1291 **Does your manuscript contain any analysis of quantitative or qualitative data?**

1292 **[X] Yes** (continue with next question)

1293 ☐ No (skip to Section 3)

1294 **Are third-party analysis scripts/code (e.g., R, Stata), codebooks, or other relevant doc-**
1295 **umentation available within a trusted digital repository?**

1296 [X] Yes, available at this link: <https://doi.org/10.17605/OSF.IO/YNG37> (~~7-R~~
1297 ~~Markdown pipeline~~ [5 analytic R Markdown reports, 2 documentation reports, shared](#)
1298 [scoring and revision-reporting](#) scripts, 4 figure scripts, 2 supplementary analysis scripts,
1299 and a Makefile; [the package is currently at OSF and prepared for planned archival](#)
1300 [migration to Zenodo](#))

1301 ☐ No, justification:

1302 Are the analysis scripts/code (e.g., R, Stata), codebooks, or other relevant documenta-
1303 tion cited in the manuscript, with a DOI?

1304 [X] Yes, the DOI is as follows: <https://doi.org/10.17605/OSF.IO/YNG37>

1305 ☐ No, justification:

1306 Section 3: Study Materials

1307 Does your manuscript contain any research materials (e.g., stimuli, programming code,
1308 questionnaires, interview protocols)?

1309 [X] Yes (continue with next question)

1310 ☐ No (skip to Section 4)

1311 Are all study materials and descriptions of study procedures available within a trusted
1312 digital repository?

1313 [X] Yes, available at this link: <https://doi.org/10.17605/OSF.IO/YNG37> (Czech-
1314 language versions of all questionnaires: HPT, FR-LF, KSA-3, SDR-5, and historical

1315 knowledge test)

1316 ☐ No, justification:

1317 **Are all third-party study materials, descriptions of study procedures, or other relevant**
1318 **documents cited in the manuscript, with a DOI?**

1319 ☐ Yes, the DOI is as follows:

1320 **[X] No, justification:** Third-party instruments (~~HPT, FR-LF, KSA-3, SDR-5~~) are cited via
1321 standard bibliography references (Hartmann & Hasselhorn, 2008; ~~Becker-Decker~~ et al.,
1322 ~~2020; Lederer, 2002; Stöber, 2001~~[2013; Beierlein et al., 2014; Hays et al., 1989](#)); DOIs are
1323 included where available. The adapted Czech versions are deposited on OSF ([https://](https://doi.org/10.17605/OSF.IO/YNG37)
1324 doi.org/10.17605/OSF.IO/YNG37).

1325 **Section 4: Preregistration**

1326 **Were any aspects of your manuscript preregistered?**

1327 **[X] Yes** (continue with next question)

1328 ☐ No (do not complete the rest of the form)

1329 **Does the manuscript contain an accessible link to the preregistration?**

1330 **[X] Yes, available at this link:** <https://osf.io/zsngy>

1331 ☐ No, justification:

1332 Do you clearly indicate in the manuscript which parts were preregistered and which
1333 parts were not?

1334 [X] Yes. (~~Preregistered components include hypotheses H1–H2, primary multilevel~~
1335 ~~models, DIF/MG-CFA analyses, and sample size determination; the TOST equivalence~~
1336 ~~analysis is explicitly labelled “exploratory” in Section 4.10 as it used a post-hoc SESOI~~
1337 ~~wider than the preregistered power target~~Section 2.5 gives a concise disclosure of
1338 post-registration changes and omitted registered analyses. Table S5 distinguishes
1339 registered tests, omissions, implemented modifications, and post-registration analyses.)

1340 ☐ No, justification:

1341 Are all preregistered analyses reported in the text or linked in the supplemental mate-
1342 rial?

1343 ☐ Yes.

1344 [X] ~~Yes.~~No, justification: (~~All preregistered hypotheses and analyses are reported in~~
1345 ~~Sections 4.1–4.10 and the supplementary materials)~~The registered continuous-ideology
1346 MIMIC model, both FR-LF-by-knowledge and KSA-by-knowledge moderation analyses
1347 and their simple slopes, clustered correlation comparison, complete covariate set,
1348 one-tailed tests, Benjamini–Hochberg correction, and registered missing-data rule were
1349 not all implemented. Table S5 reports these omissions and the analyses used instead.

1350 ~~—☐ No, justification:~~

1351 Are all deviations from the preregistration plan clearly disclosed in the manuscript

1352 (either in text or in a table)?

1353 [X] Yes. (~~One deviation: TOST-SESOI set post hoc at~~ Section 2.5 and Table S5 disclose the
1354 changed inferential hierarchy, omitted registered procedures, instrument discrepancies,
1355 and post-registration additions. TOST and its ± 0.20 ~~instead of the preregistered power~~
1356 ~~target of $\beta \geq .15$; disclosed in Section 4.10 and the analysis is labelled exploratory~~ SESOI
1357 are explicitly labelled exploratory.)